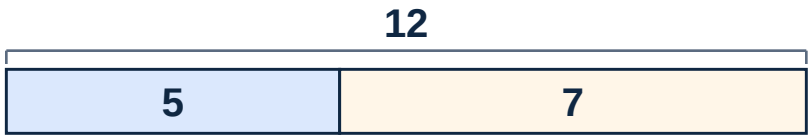
# One-Step Equations

Find the number that makes it true.

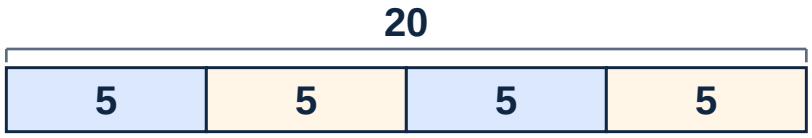
### x + 5 = 12: the bar shows the whole




12 is the whole. 5 is one part.
The other part is 12 − 5.

**x = 7**

### 4x = 20: four equal parts make 20




4x means 4 times x.
Each part is 20 ÷ 4.

**x = 5**

### Check it: put the number back in

**x = 7: 7 + 5 = 12 true!**

**Your turn!**

**x + 9 = 23: x =** 14

**For teachers**

Goal: solve one-step equations x + p = q and px = q, check a solution, and write an equation for a story (6.EE.5–7). Say “What is the whole? What is the part?” Draw the bar first.

# One-Step Equations: Practice

Draw the bar. Then check your answer.

**Part 1** Is it true? Put the number in for x. Circle yes or no.

1) **x = 4 in x + 7 = 11** — (yes) / no

2) **x = 6 in 3x = 21** — yes / (no)

3) **x = 9 in x + 8 = 17** — (yes) / no

4) **x = 5 in 4x = 20** — (yes) / no

5) **x = 12 in x + 3 = 16** — yes / (no)

6) **x = 8 in 6x = 42** — yes / (no)

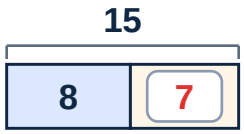
7) **x = 7 in x + 13 = 20** — (yes) / no

8) **x = 3 in 9x = 30** — yes / (no)

## One-Step Equations: Practice (continued)

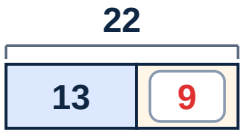
**Part 2** Find the missing part The bar shows the whole.

9) $x + 8 = 15$



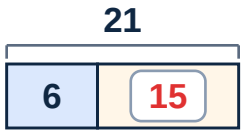

$x =$ 7

10) $x + 13 = 22$




$x =$ 9

11) $x + 6 = 21$



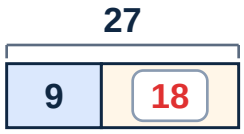

$x =$ 15

12) $x + 24 = 35$




$x =$ 11

13) $x + 9 = 27$



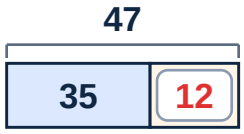

$x =$ 18

14) $x + 17 = 23$



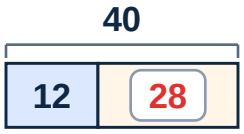

$x =$ 6

15) $x + 35 = 47$




$x =$ 12

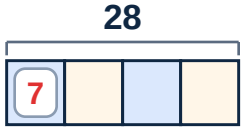
16) $x + 12 = 40$




$x =$ 28

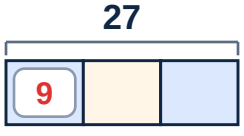
**Part 3** Equal parts Find one part.

17) $4x = 28$




$x =$ 7

18) $3x = 27$

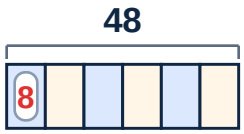



$x =$ 9

## One-Step Equations: Practice (continued)

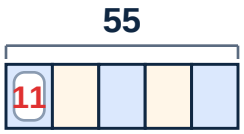
### Part 3 Equal parts (continued)

19) $6x = 48$



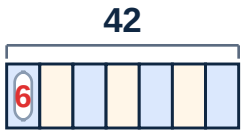

$x =$ 8

20) $5x = 55$



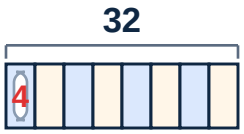

$x =$ 11

21) $7x = 42$



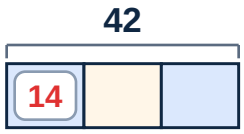

$x =$ 6

22) $8x = 32$



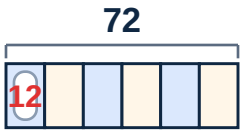

$x =$ 4

23) $3x = 42$




$x =$ 14

24) $6x = 72$




$x =$ 12

### Part 4 Plus equations No help this time.

25) $x + 13 = 41$: $x =$ 28

26) $x + 27 = 42$: $x =$ 15

27) $x + 9 = 55$: $x =$ 46

28) $x + 35 = 72$: $x =$ 37

29) $x + 18 = 72$: $x =$ 54

30) $x + 46 = 65$: $x =$ 19

## One-Step Equations: Practice (continued)

### Part 4 Plus equations (continued)

31) $x + 7 = 75$: $x =$ **68**

32) $x + 52 = 85$: $x =$ **33**

33) $x + 24 = 73$: $x =$ **49**

34) $x + 61 = 86$: $x =$ **25**

35) $x + 16 = 92$: $x =$ **76**

36) $x + 39 = 81$: $x =$ **42**

### Part 5 Times equations No help this time.

37) $6x = 54$: $x =$ **9**

38) $8x = 56$: $x =$ **7**

39) $4x = 52$: $x =$ **13**

40) $9x = 72$: $x =$ **8**

41) $7x = 84$: $x =$ **12**

42) $5x = 80$: $x =$ **16**

## One-Step Equations: Practice (continued)

### Part 5 Times equations (continued)

43) $12x = 72$: $x =$ **6**

44) $3x = 81$: $x =$ **27**

45) $11x = 55$: $x =$ **5**

46) $8x = 120$: $x =$ **15**

47) $6x = 84$: $x =$ **14**

48) $10x = 230$: $x =$ **23**

### Part 6 More plus equations

49) $x + 35 = 94$: $x =$ **59**

50) $x + 15 = 58$: $x =$ **43**

51) $x + 33 = 61$: $x =$ **28**

52) $x + 55 = 94$: $x =$ **39**

53) $x + 20 = 49$: $x =$ **29**

54) $x + 14 = 41$: $x =$ **27**

## One-Step Equations: Practice (continued)

### Part 6 More plus equations (continued)

55) $x + 18 = 87$: $x =$ **69**

56) $x + 28 = 67$: $x =$ **39**

57) $x + 76 = 128$: $x =$ **52**

58) $x + 52 = 121$: $x =$ **69**

59) $x + 53 = 110$: $x =$ **57**

60) $x + 61 = 97$: $x =$ **36**

61) $x + 42 = 102$: $x =$ **60**

62) $x + 67 = 144$: $x =$ **77**

63) $x + 28 = 88$: $x =$ **60**

64) $x + 36 = 114$: $x =$ **78**

65) $x + 60 = 69$: $x =$ **9**

66) $x + 20 = 55$: $x =$ **35**

## One-Step Equations: Practice (continued)

### Part 7 More times equations

67) $11x = 44$: $x =$ **4**

68) $12x = 228$: $x =$ **19**

69) $5x = 100$: $x =$ **20**

70) $8x = 152$: $x =$ **19**

71) $12x = 168$: $x =$ **14**

72) $4x = 84$: $x =$ **21**

73) $3x = 6$: $x =$ **2**

74) $11x = 231$: $x =$ **21**

75) $2x = 6$: $x =$ **3**

76) $3x = 36$: $x =$ **12**

77) $12x = 96$: $x =$ **8**

78) $7x = 126$: $x =$ **18**

## One-Step Equations: Practice (continued)

### Part 7 More times equations (continued)

79) $10x = 110$: $x =$ **11**

80) $3x = 72$: $x =$ **24**

81) $2x = 48$: $x =$ **24**

82) $4x = 80$: $x =$ **20**

83) $7x = 91$: $x =$ **13**

84) $12x = 24$: $x =$ **2**

### Part 8 Remember? Problems from earlier packets.

85) $-27$ **>** $-31$

86) opposite of $-45 =$ **45**

87) LCM of 9 and 15 = **45**

88) when $n = 7$: $4n + 6 =$ **34**

89) $8.75 \div 0.25 =$ **35**

90) 35% of 80 = **28**

## One-Step Equations: Practice (continued)

**Part 9** Word problems Write the equation. Then solve it.

91) Leo has some cards. He gets 14 more and now has 51. Write an equation and find x.

Number sentence x + 14 = 51, x = 37

Answer 37 cards

Draw or show your work

92) 6 friends share the cost of a gift equally. Each pays $9. What does the gift cost?

Number sentence x ÷ 6 = 9, x = 54

Answer $54

Draw or show your work

93) A box of 8 equal bags weighs 96 ounces. How much does each bag weigh?

Number sentence 8x = 96, x = 12

Answer 12 ounces

Draw or show your work

94) Ava read x pages. Then she read 25 more, 63 pages in all. How many pages first?

Number sentence x + 25 = 63, x = 38

Answer 38 pages

Draw or show your work

95) Challenge: 3 packs of pens cost $27 and each pack has 4 pens. What does 1 pen cost?

Number sentence 3x = 27, x = 9, 9 ÷ 4 = 2.25

Answer $2.25

Draw or show your work

# Check-In: One-Step Equations

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

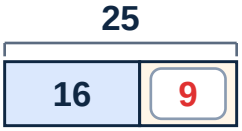
**Part 1** Is it true? Circle yes or no.

1) $x = 8$ in $7x = 56$ — (yes) / no

2) $x = 7$ in $5x = 30$ — yes / (no)

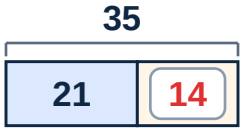
**Part 2** Find the missing part

3) $x + 16 = 25$




$x =$ 9

4) $x + 21 = 35$




$x =$ 14

**Part 3** Solve

5) $7x = 63$: $x =$ 9

6) $4x = 64$: $x =$ 16

7) $x + 14 = 53$: $x =$ 39

8) $x + 8 = 65$: $x =$ 57